

A. Compact system overview

Print proof: 5.5-inch figure width, 10-point body text.

Building systems that collaborate effectively with people is a longstanding goal of interactive machine learning. Recent advances in LLM agents have enabled collaboration on digital tasks through dialogue, preference adaptation, and coordinated action. Visual assistants extend this collaboration to everyday physical activities, observing the user’s ongoing task and providing guidance.

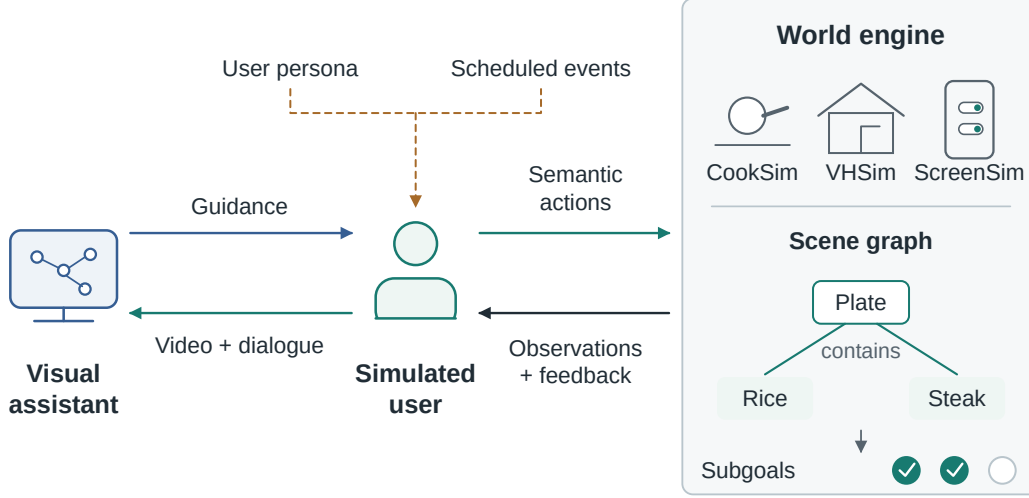


Figure 1. Interactive Visual Gym. The assistant communicates with a simulated user, who alone executes semantic actions in the world engine. The engine supports three task domains and tracks subgoal completion from its scene-graph state. Icons denote domains; the graph is an illustrative partial cooking state.

Action interface. Only the simulated user acts on the environment. The engine executes semantic actions, updates its internal state, and renders the observations available to the user.

Communication interface. The visual assistant receives the user-view stream and dialogue, and influences subsequent execution through guidance. User personas and task-state-triggered events vary the interaction.

B. Environment-first overview

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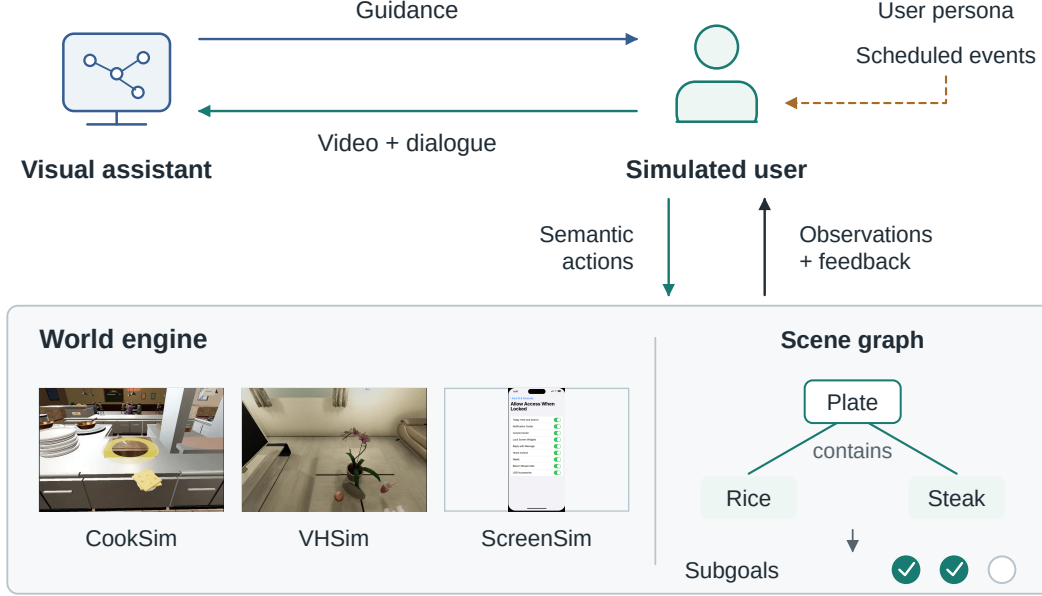


Figure 1. Interactive Visual Gym. Communication and execution form two connected feedback loops. The expanded world engine shows authentic simulator views and an illustrative scene graph used for subgoal checks. The assistant receives rendered user-view observations, not the internal graph.

Action interface. Only the simulated user acts on the environment. The engine executes semantic actions, updates its internal state, and renders the observations available to the user.

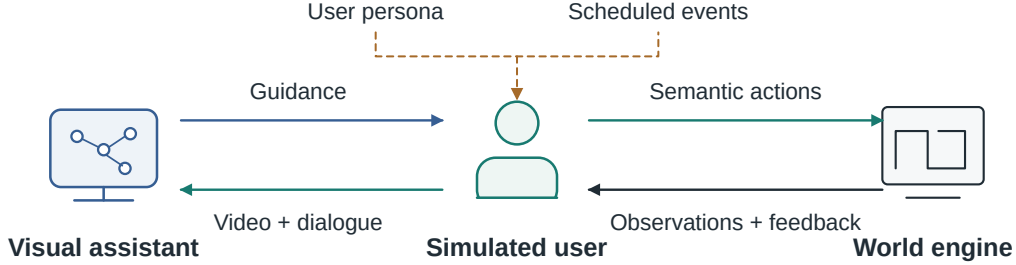
Communication interface. The visual assistant receives the user-view stream and dialogue, and influences subsequent execution through guidance. User personas and task-state-triggered events vary the interaction.

C. Interaction and engine detail

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a Closed-loop assistance



b Executable task environments

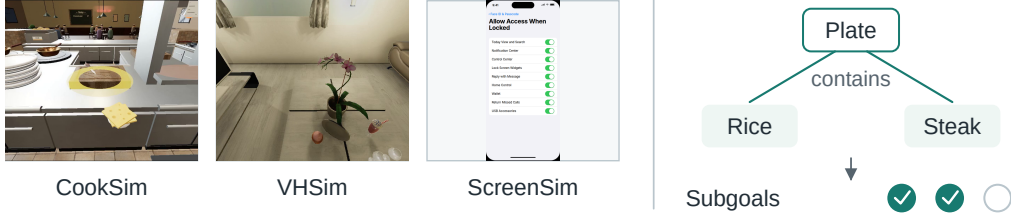


Figure 1. Interactive Visual Gym. (a) Only the simulated user acts on the world; the assistant communicates through guidance. Personas and human events condition user behavior. (b) Three executable task environments share semantic action interfaces and state-based progress tracking; the graph illustrates a partial cooking state.

Action interface. Only the simulated user acts on the environment. The engine executes semantic actions, updates its internal state, and renders the observations available to the user.

Communication interface. The visual assistant receives the user-view stream and dialogue, and influences subsequent execution through guidance. User personas and task-state-triggered events vary the interaction.

D. Minimal architecture

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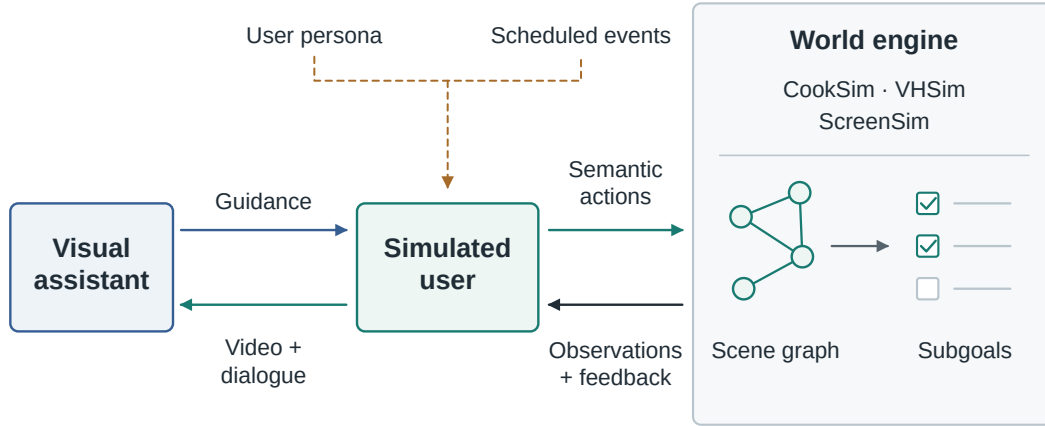


Figure 1. Interactive Visual Gym. The assistant influences task execution through communication with the simulated user, who alone acts in the world engine. User personas and scheduled events condition the user. The three task environments maintain scene-graph state to determine subgoal completion.

Action interface. Only the simulated user acts on the environment. The engine executes semantic actions, updates its internal state, and renders the observations available to the user.

Communication interface. The visual assistant receives the user-view stream and dialogue, and influences subsequent execution through guidance. User personas and task-state-triggered events vary the interaction.